

Development of a Deep Learning Pipeline for Automated Plant Disease Diagnosis Using Sequential CNNs

by

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1.0 PROJECT OVERVIEW

Plant diseases are one of the major factors affecting agricultural productivity and crop quality worldwide. Traditional methods of identifying plant diseases usually depend on manual observation by farmers or agricultural experts, which can be time-consuming, costly, and sometimes inaccurate. With the continuous advancement of Artificial Intelligence (AI) and Deep Learning technologies, automated image classification systems have become effective tools for detecting plant diseases through digital images. Convolutional Neural Networks (CNN) are widely used in image recognition tasks because of their ability to automatically identify patterns and features within images. This project proposes a Plant Disease Recognition System using CNN with TensorFlow and KerasCV that can analyze plant leaf images and determine possible diseases efficiently and accurately. The system aims to provide users with a faster and more accessible method of plant disease detection through image processing and machine learning.

01 | Problem Statement

Plant diseases can significantly reduce crop yield and negatively affect agricultural production when not detected early. Many farmers still rely on traditional manual inspection methods, which may lead to delayed or inaccurate diagnosis due to limited expertise or resources. In addition, access to agricultural specialists may not always be available, especially in remote areas. Because of these limitations, there is a need for an automated and intelligent system capable of identifying plant diseases using image-based analysis. The proposed system addresses this problem by utilizing Convolutional Neural Networks (CNN) with TensorFlow and KerasCV to recognize diseases from plant leaf images and provide immediate diagnostic results.

02 | Target users and their needs

The primary target users of the system are farmers, agriculture students, researchers, and individuals involved in plant cultivation. Farmers need a fast and reliable way to identify plant diseases in order to take immediate action and prevent crop damage. Agriculture students and researchers require a system that can assist in studying plant diseases and machine learning applications in agriculture. Users also need a convenient and accessible platform that allows them to upload plant leaf images and receive accurate disease predictions without requiring advanced technical knowledge. The system is designed to meet these needs by providing an easy-to-use interface and automated disease recognition functionality.

03 | Key requirements

The proposed Plant Disease Recognition System requires both functional and non-functional requirements to ensure proper operation and performance. The system must allow users to upload plant leaf images for analysis and display prediction results generated by the CNN model. It must also provide disease information and possible treatment recommendations based on the detected condition. The system should support image preprocessing and classification using TensorFlow and KerasCV. In addition, the interface must be user-friendly, responsive, and capable of processing images efficiently. The system should also maintain reliable prediction accuracy and support dataset management and model training functionalities for administrators.

04 | Objectives

The main objective of this project is to develop a Plant Disease Recognition System using Convolutional Neural Networks (CNN) with TensorFlow and KerasCV that can accurately identify plant diseases through image classification of plant leaf images. Specifically, this project aims to: (1) implement a Convolutional Neural Network (CNN) model using TensorFlow and KerasCV for detecting and classifying plant diseases from uploaded leaf images; (2) design and develop a user-friendly interface that allows users to upload plant leaf images and view diagnostic results efficiently; and (3) evaluate the performance and accuracy of the proposed system in identifying plant diseases using image-based analysis.

2.0 USE CASE DIAGRAMS

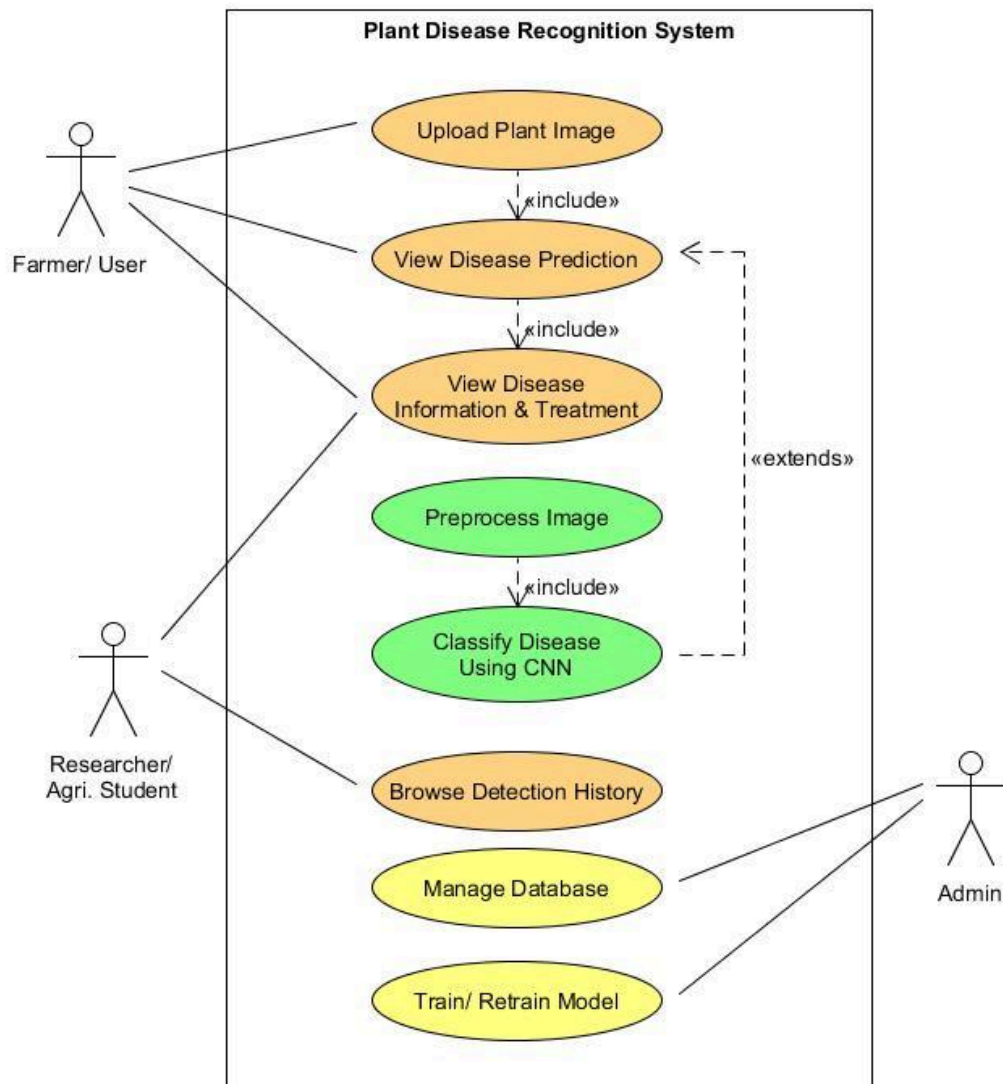


Figure 1. Use Case Diagram for Plant Disease Detection System

Figure 1 illustrates the interactions between the users and the Plant Disease Detection System. The main actors in the system are the Farmer, Researcher or Agriculture Student, and Admin. The Farmer can upload leaf images, view prediction results, and view treatment recommendations, while the Admin is responsible for managing the database and training the AI model. Furthermore, the researcher or agriculture student can also view disease information, treatment, and browse detection history. The system is responsible for

processing the uploaded image and detecting possible plant diseases using image analysis, particularly Tensorflow and KerasCV. The diagram also shows the relationships (<<include>>), where uploading an image includes image processing, and image processing includes disease detection.

Item	Description
Use Case Name	Upload Leaf Image
Actor	Farmer/ User (Primary); Agriculture Student/ Researcher (Secondary)
Description	The user accesses the user interface and submits a digital image of a plant leaf for disease analysis. The system accepts the uploaded image, validates its format and quality, then queues it for image processing and CNN-based classification.
Pre-condition	• The user has access to the system (via web or mobile interface). • A plant leaf image is available in a supported format (JPG, PNG). • The system is online and the CNN model is loaded and ready.
Post-condition	• The image is successfully received and stored in the database • Image preprocessing is automatically triggered

Table 1. Key Use Case 1: Upload Leaf Image

Table 1 shows the use case, which allows the farmer to upload an image of a plant leaf into the system. After the image is uploaded, the system automatically processes the image for disease analysis, which detects whether it is healthy or has a disease.

Item	Description
Use Case Name	View Diagnosis Result
Actor	Farmer/ User (Primary); Agriculture Student/ Researcher (Secondary)
Description	The user views the system-generated result after a plant leaf image has been analyzed, showing whether the plant is healthy or diseased and identifying the possible disease.
Pre-condition	The user has already uploaded a plant leaf image and the system has completed image processing and analysis.
Post-condition	The diagnosis result is successfully displayed to the user, and the system stores or retains the result for reference (if applicable).

Table 2. Key Use Case 2: View Diagnosis Result

Table 2 shows the table to view the result generated by the system after the plant leaf image has been analyzed. The system displays whether the plant is healthy or infected with a disease. This allows the user to understand the condition of the plant based on the image input, and the result allows the user to decide on possible actions for plant treatment.

CLASS DIAGRAMS

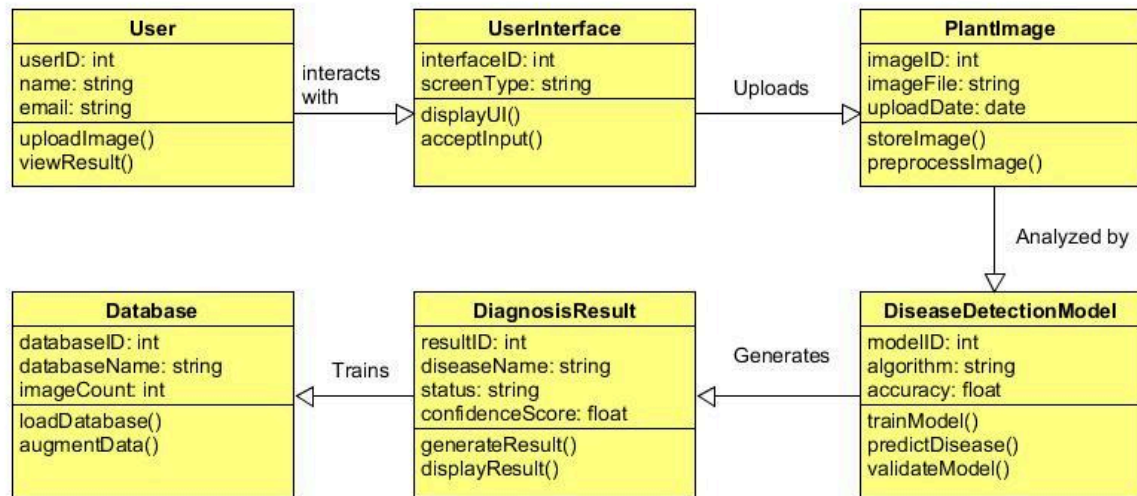


Figure 2. Class Diagram for Plant Disease Detection System

Figure 2 shows the class diagram that illustrates the relationships and structure of the Plant Disease Detection System. The diagram consists of 6 classes, namely: User, UserInterface, PlantImage, DiseaseDetectionModel, DiagnosisResult, and Database. The user interacts with the system interface to upload plant images, while the CNN model is used to process the image from the database. After the processing stage, the diagram shows that the diagnosis results are displayed and generated. Overall, this diagram represents the relationships between the different classes with its attributes and methods.

ANALYSIS

01 | Methodology

This project implements Convolutional Neural Network (CNN) to automate plant disease recognition. Utilizing the TensorFlow environment backed with Keras utilities for dataset structuring and neural network layers. The dataset will be streamed into the training environment using core TensorFlow preprocessing functions. Specifically, `tf.keras.utils.image_dataset_from_directory()`. The primary backbone of the classification system relies on a Convolutional Neural Network (CNN). The architecture is constructed using a sequential arrangement of structural Keras layers (`tf.keras.layers.Conv2D`, `MaxPooling2D`, and `Dense`).

02 | Data Collection

The dataset used for this project is [New Plant Diseases Dataset](#) from kaggle. During the preprocessing, we will be using the built-in function of KerasCV `tf.keras.utils.image_dataset_from_directory()` for both the training set and test set. This will ensure that our dataset will be formatted before building the model/layers for the Convolutional Neural Network (CNN).

03 | Model Development

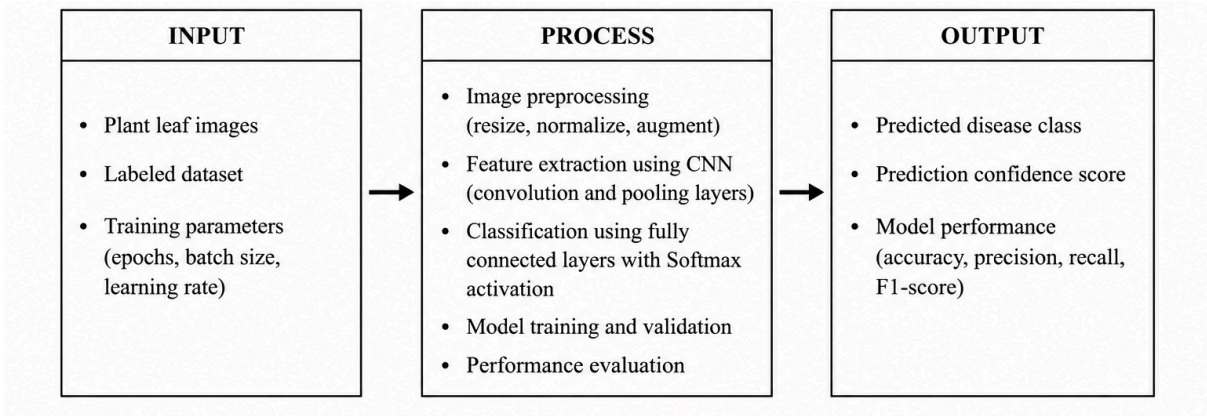


Figure 3. CNN-Based Plant Disease Recognition System Framework

Figure 3 illustrates the overall workflow and Input-Process-Output (IPO) framework of the proposed CNN-based Plant Disease Recognition System, where input leaf images from the dataset, along with training parameters, first undergo preprocessing steps such as resizing, normalization, and data augmentation to improve consistency and generalization. The processed images are then fed into a Convolutional Neural Network (CNN) implemented using TensorFlow and KerasCV, where feature extraction is performed through convolutional layers, dimensionality reduction through pooling layers, and final classification through fully connected dense layers, with ReLU activation used in hidden layers and Softmax activation used for multi-class prediction. The dataset is split into training and validation sets to allow the model to learn patterns while being evaluated on unseen data, using the Adam optimizer and categorical cross-entropy loss function, with performance monitored through validation accuracy and validation loss across epochs and optional early stopping to prevent overfitting. Finally, the system outputs the predicted disease class, confidence score, and evaluation metrics such as accuracy, precision, recall, and F1-score, providing a complete and reliable plant disease detection result.

04 | Software Interface

Abc

05 | Model Performance Evaluation

The performance of the Plant Disease Recognition System will be evaluated to measure the effectiveness of the CNN model in classifying plant leaf diseases and its ability to generalize to unseen validation and test data. This ensures that the model is reliable and suitable for real-world use.

The evaluation metrics include accuracy, precision, recall, and F1-score. Accuracy measures overall correct predictions, precision measures the correctness of predicted positive classes, recall measures the ability to identify actual positive cases, and F1-score provides a balanced measure of precision and recall, which is especially useful for imbalanced datasets. For multi-class classification, the F1-score will be computed using macro-averaging so that each class contributes equally to the final result.

A confusion matrix will also be used to analyze the classification results by showing the relationship between actual and predicted classes.

Predicted \ Actual	Healthy	Disease A	Disease B
Healthy	X_{11}	X_{12}	X_{13}
Disease A	X_{21}	X_{22}	X_{23}
Disease B	X_{31}	X_{32}	X_{33}

Figure 4. Confusion Matrix for Plant Disease Classification Results

Where X_{ij} represents the number of samples for each actual-predicted class pair, with diagonal values indicating correct predictions and off-diagonal values indicating misclassifications.

Training and validation accuracy and loss will also be monitored to evaluate model learning behavior and check for overfitting or underfitting. This helps ensure that the model generalizes well to unseen plant leaf images.

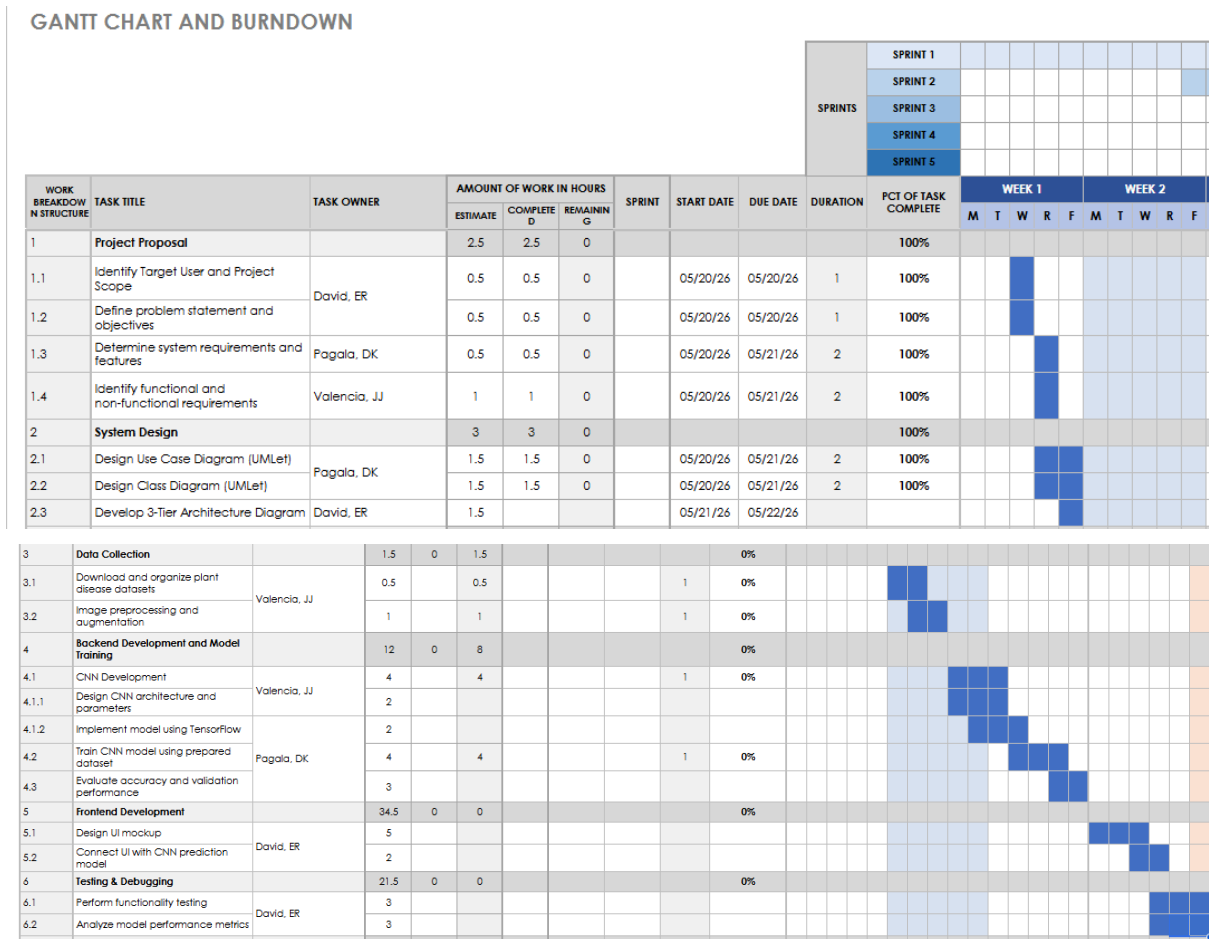
06 | Expected Outcomes

The proposed project on Plant Disease Detection aims to provide an efficient and accurate method for identifying diseases using image classification through CNN. The system allows the user to upload single or multiple files of plant leaf images, which will also produce a diagnostic result whether the plant is healthy or infected. Furthermore, this system aims to improve the speed and convenience of plant disease detection compared to traditional manual inspection methods. Overall, this proposed project will help the students apply their learnings on the application of CNN model that will use Tensorflow and KerasCV.

APPENDICES

01 | Project Timeline

Link: [CPE178P_E01_3T2425_Project_Sked_Group03](#)



02 | GitHub Repository

Link: [CPE178P_E01_3T2425_GitHub_Group03](#)